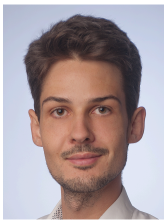
Vincent **Bürgin**

vincent@vincent-buergin.de   https://vincent-buergin.de
Vuenc   linkedin.com/in/vincent-buergin



Education

Ph.D. Student in Computer Science — Technical University of Munich (TUM)
Supervisor: Prof. Stefanie Jegelka — June 2025 – present

- Ph.D. Student (MCML & ELLIS) at Chair for *Foundations of Deep Neural Networks* (Prof. Stefanie Jegelka)

M.Sc. Informatics — Technical University of Munich (TUM)
Grade: 1,0 / *passed with high distinction* — October 2020 – March 2024

- Master's Thesis: *Topology-Aware 3D Medical Image Segmentation via Persistent Homology* (Grade: 1,0)
  Supervision: Prof. Daniel Rückert, Prof. Ulrich Bauer, Dr. Johannes Paetzold, Nico Stucki
- Area of Specialization: *Machine Learning and Analytics*

B.Sc. Computer Science — University of Passau
Grade: 1,0 / *mit Auszeichnung ("with distinction")* — October 2016 – September 2020

- Bachelor's Thesis: *Distribution-Valued Games: Overview, Analysis, and a Segmentation-Based Approach* (Grade: 1,0)
  Supervision: Prof. Fabian Wirth, Prof. Hermann de Meer, Ali Alshawish

B.Sc. Mathematics — University of Passau
Grade: 1,0 / *mit Auszeichnung ("with distinction")* — April 2019 – September 2020

- Studied jointly with B.Sc. Computer Science, joint Bachelor's Thesis

Abitur *(general university entrance qualification)* — Goethe-Gymnasium Regensburg
Grade: 1,0 — September 2008 – June 2016

Work Experience

Engineering Internship — QuantCo, Munich
January 2025 – May 2025

- Cloud deployment of inference server *(Kubernetes, AWS EKS, Terraform)*, performance testing, observability (*OpenTelemetry*)

Research Internship — ImFusion GmbH, Munich
November 2022 – April 2023

- Researching graph neural networks (GNNs) for vertebra identification on CT scans *(Python, C++, PyTorch Geometric)*
- Research resulted in publication at MICCAI conference (*Robust vertebra identification using simultaneous node and edge predicting Graph Neural Networks*, MICCAI 2023).

Student assistant — Chair of Computer Networks, University of Passau
March 2019 – December 2019

- Integrating the *ns-3* network simulator with the mosaik smart power grid co-simulation framework *(Python)*

Volunteering

EnHands — October 2022 – present
Non-profit association and student initiative at TUM developing hand prosthesis prototypes for low-income countries

- Founding member, leading marketing/public relations team (2023 – present), responsible for finances (treasurer of EnHands e.V. association, 2024 – present), co-organizing new member onboarding (2022 – present)

Academic Publications

7. **Beyond Structural Symmetries: Linear Mode Connectivity via Neuron Identifiability.**
   V. Bürgin*, D. Herbst*, Y.-W. E. Lin, S. Jegelka (2026). Forty-third International Conference on Machine Learning (ICML), 2026.
6. **Efficient Betti Matching Enables Topology-Aware 3D Segmentation via Persistent Homology.**
   N. Stucki, V. Bürgin, J. C. Paetzold, and U. Bauer (2024). Preprint, arXiv:2407.04683.
5. **Topologically faithful multi-class segmentation in medical images.**
   A. Berger, N. Stucki, L. Lux, V. Bürgin, S. Shit, A. Banaszak, D. Rückert, U. Bauer, and J. C. Paetzold (2024). Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024.
4. **On the Localization of Ultrasound Image Slices within Point Distribution Models.**
   L. Bastian*, V. Bürgin*, H. Y. Kim*, A. Baumann, B. Busam, M. Saleh, and N. Navab (2023). International Workshop on Shape in Medical Imaging (ShapeMI), 2023.
3. **S3M: Scalable Statistical Shape Modeling through Unsupervised Correspondences.**
   L. Bastian*, A. Baumann*, E. Hoppe, V. Bürgin, H. Y. Kim, M. Saleh, B. Busam, and N. Navab (2023). Medical Image Computing and Computer Assisted Intervention (MICCAI) 2023.
2. **Robust vertebra identification using simultaneous node and edge predicting Graph Neural Networks.**
   V. Bürgin, R. Prevost, and M. F. Stollenga (2023). Medical Image Computing and Computer Assisted Intervention (MICCAI) 2023.
1. **Remarks on the tail order on moment sequences.**
   V. Bürgin, J. Epperlein, and F. Wirth (2022). Journal of Mathematical Analysis and Applications, Vol. 512 (1).

* Denotes equal contributions

Scholarships and Achievements

ELLIS: European Laboratory for Learning and Intelligent Systems — 2025 – present
Ph.D. Student in ELLIS academic track / Co-Supervisor: Prof. Martin Jaggi (EPFL)

relAI: Konrad Zuse School of Excellence in Reliable AI — October 2022 – September 2024
Scholarship and program for Master's and Ph.D. students focusing on reliable AI

best.in.tum, Technical University of Munich — April 2022
Award for top 2% students of the TUM Department of Informatics

Faculty Prize, University of Passau — May 2022
Award for the two highest-scoring graduates per degree program / received for both my B.Sc. degrees

Max Weber-Programm — 2016 – 2022
Merit-based scholarship, awarded after highschool and confirmed after two years of studies

Bundeswettbewerb Informatik/Bundeswettbewerb Mathematik — 2013 – 2016
National highschool student CS/math competitions. Participated 3× (BWINF; final round participant in 2016)/2× (BWM)

Skills

Language Skills

- **German**: Native speaker
- **English**: Fluent and experienced speaker (C1–C2) / TOEFL score 118/120 (June 2020)
- **Spanish**: Basic to advanced knowledge (B1–B2) / university courses until B2.1 level
- **French**: Basic knowledge

Programming Languages and Software

- **Python**: Extensive experience through research projects, research internship, university projects and personal projects. Libraries used include *Numpy*, *SciPy*, *Pandas*, *PyTorch*, *PyTorch Geometric*.
- **C++**: Experience through research internship and Master's thesis project.
- **Rust**: Experience through personal projects.
- **JavaScript/NodeJS**: Experience through university projects (full-stack web development) and personal projects (interactive visualizations and algorithmic art). Libraries used include *VueJS*, *Express*, *p5.js*.
- **C#**: Experience through personal projects (2010 – 2016) and highschool student jobs.

Other technologies: *Git* / *Java*, *C*, *Haskell*, *Julia* / *LaTeX*, *Typst* / *AWS*, *Kubernetes* / *SageMath*, *Mathematica*, *MATLAB*, *R*.